Topic 9 — Pollution & Waste Management

Covers syllabus (click to expand)

Environmental Pollution (पर्यावरणीय प्रदूषण) | Environmental Degradation (पर्यावरणीय क्षरण) | Air Pollution | Water Pollution | Soil Pollution | Noise Pollution | Radioactive Pollution | Thermal Pollution | Marine Pollution | Plastic Pollution | Solid Waste Management | Biomedical Waste | Hazardous Waste | E-waste

Sources baked in: Water Act 1974, Air Act 1981, EPA 1986, SWM/BMW/HW/E-waste Rules, CPCB/CGWB, NCAP 2019, UPPCS PYQs 2018–2025 **Weight:** ★★★ High — standards, org↔report traps, UP rivers/cities **Last verified:** August 2026

Current Affairs (this topic)

- NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) / CAQM / solid-waste and e-waste rule updates are the living pollution CA layer.
- Keep primary pollutant vs secondary pollutant and BOD vs COD pairs stable.

9.1 Environmental Pollution

Identity: Beyond absorption capacity | Point/non-point | Primary/secondary | Water–Air–EPA laws

- **Environmental pollution** is the harmful addition of substances or energy when the rate exceeds the environment's **absorption or assimilative capacity**.
- Pollutants may be **physical** (heat, noise, radiation), **chemical** (metals, pesticides, gases), or **biological** (pathogens).
- Pollution caused or strongly influenced by humans is called **anthropogenic** pollution; the human-modified environment is the **anthropogenic environment**.
- **Biodegradable** pollutants (sewage, faecal matter, urine, domestic organic waste) can be broken down by microbes. **Non-biodegradable** pollutants (pesticides, plastics, many metals, asbestos) persist or degrade very slowly.
- **Point sources** have identifiable outlets such as factory (फैक्टरी) chimneys and sewage pipes.
- **Non-point sources** are diffuse, such as fertilizer runoff from farms and urban (नगरीय) stormwater.
- **Primary pollutants** are emitted directly, including **SO_2**, **CO**, **NOx**, and **particulate matter**.
- **Secondary pollutants** form in the atmosphere, including **ground-level ozone**, **PAN**, photochemical smog, and **acid rain**.
- The **Water (Prevention and Control of Pollution) Act, 1974** controls water pollution.
- The **Air (Prevention and Control of Pollution) Act, 1981** controls air pollution (noise included by **1987** amendment).

- The **Environment (Protection) Act, 1986** is the umbrella law for hazardous substances, standards, and EIA.
- **CPCB** sets national standards. **SPCBs** enforce pollution control at the state level.

> **Logic:** Pollution requires exceeding tolerance — not every human addition counts. Pesticide ≠ biodegradable.

PYQ — UPPCS Prelims 2025, Q22

Carrying capacity: (1) Resource extraction should remain above regeneration. (2) Waste should stay within absorption capacity.

A. Only 2

B. Neither

C. Both

D. Only 1

Show answer

Ans: A — Extraction must be **at or below** regeneration.

9.2 Environmental Degradation

Identity: Broader than pollution | Carrying capacity | Deforestation–mining–urbanization chains

- **Environmental degradation** is the decline in environmental quality and productivity from pollution, resource depletion, and ecosystem (पारिस्थितिकी तंत्र) damage.
- Degradation is **broader than pollution**. Deforestation without toxic chemicals still causes degradation.
- A common chain runs from **deforestation** → **soil erosion** → **river siltation** → **flood risk** → **productivity loss**.
- **Urbanization** degrades the environment through sewage overload, heat islands, and waste accumulation.
- **Mining** removes vegetation and can cause acid drainage and toxic tailings. **Sonbhadra (सोनभद्र) (UP)** faces mining-related degradation.
- **Intensive agriculture** degrades soil through pesticide buildup, salinization, and organic matter loss.
- Degradation accelerates when **resource extraction exceeds regeneration** or **waste exceeds absorption capacity**.
- About **30% of India's land area** is degraded according to ISRO desertification (मरुस्थलीकरण) assessments.
- **EIA under EPA 1986** aims to prevent degradation before major projects start.

> **Logic:** Afforestation **reverses** degradation; it is not a cause of it.

9.3 Air Pollution

Identity: PM2.5/PM10 | Smog types | AQI | NCAP | Lichens as bio-indicators

- **Air pollution** is the presence of harmful gases, dust, and smoke at levels that injure health or visibility.
- **Natural sources** include volcanoes and forest fires. **Man-made sources** include fuel combustion, vehicles, and industry.
- **PM2.5** particles (≤2.5 μm) penetrate deep into the lungs and bloodstream; rain cleans coarser dust better than ultrafines.
- **PM10** particles (≤10 μm) mainly affect the upper respiratory tract.
- **SO_2** from coal combustion causes respiratory illness, acid rain, and building damage.
- **NOx** from vehicles and power plants helps form **photochemical smog** and **ground-level ozone**.
- **CO** from incomplete combustion binds haemoglobin **~200–300×** more strongly than oxygen (carboxyhaemoglobin) and can kill at high saturation.
- **Lead** from petrol antiknock additives harms children's brain development and adult heart/respiratory health.
- **London/classical smog** forms from **SO_2 and smoke** in cool, humid conditions.
- **Los Angeles/photochemical smog** forms when **NOx + hydrocarbons** react in sunlight to yield **O_3, formaldehyde, and PAN**; **PAN** is a strong eye irritant.
- India's **AQI** is based on **eight** pollutants: **PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb — not CO_2 or CH_4**. Bands: **0–50 Good … 401–500 Severe**. **NAQI** launched **17 Oct 2014**.
- **NCAP (National Clean Air Programme, 2019)** targets non-attainment cities for about **20–30% PM reduction** (2017 baseline framing).
- **Fly ash** from coal TPPs is used in bricks and as Portland cement replacement; main oxides are SiO_2, Al_2O_3, CaO, plus toxic metals (Pb, As, Cu).
- Control devices: **bag filters** for finer particulates (**<~50 μm**); **cyclone separator/collector** and wet scrubbers for coarser loads; **cyclone divider** is not a standard device name.
- **Catalytic converters** (Pt–Pd–Rh) convert CO → CO_2, NOx → N_2, and unburnt HC → CO_2 + H_2O.
- **Lichens** are sensitive **bio-indicators of air pollution**, especially **SO_2**. They **do not grow well in polluted areas**.
- **Radon** (from radium/soil) is the classic major **indoor** air pollutant and raises lung-cancer risk in poorly ventilated homes.
- **Asian Brown Cloud** aerosols are especially noted over **South Asia** (winter; black carbon, sulphate).
- **WAYU** (Wind Augmentation Purifying Unit) was installed at Delhi traffic junctions. **CCAC** (2012, UNEP partners) targets short-lived climate pollutants: **methane, black carbon, HFCs** — not a G20-only club.
- **Ringelmann scale** (रिंगेलमैन पैमाना) grades smoke density/opacity.

> **Logic:** — lichens indicate pollution but **decline** in polluted air. AQI ≠ greenhouse-gas index.

PYQ — UPPCS Prelims 2025, Q61

1. Lichens are very good indicators of pollution. 2. Lichens grow well in polluted areas.

A. Only 2

B. Neither

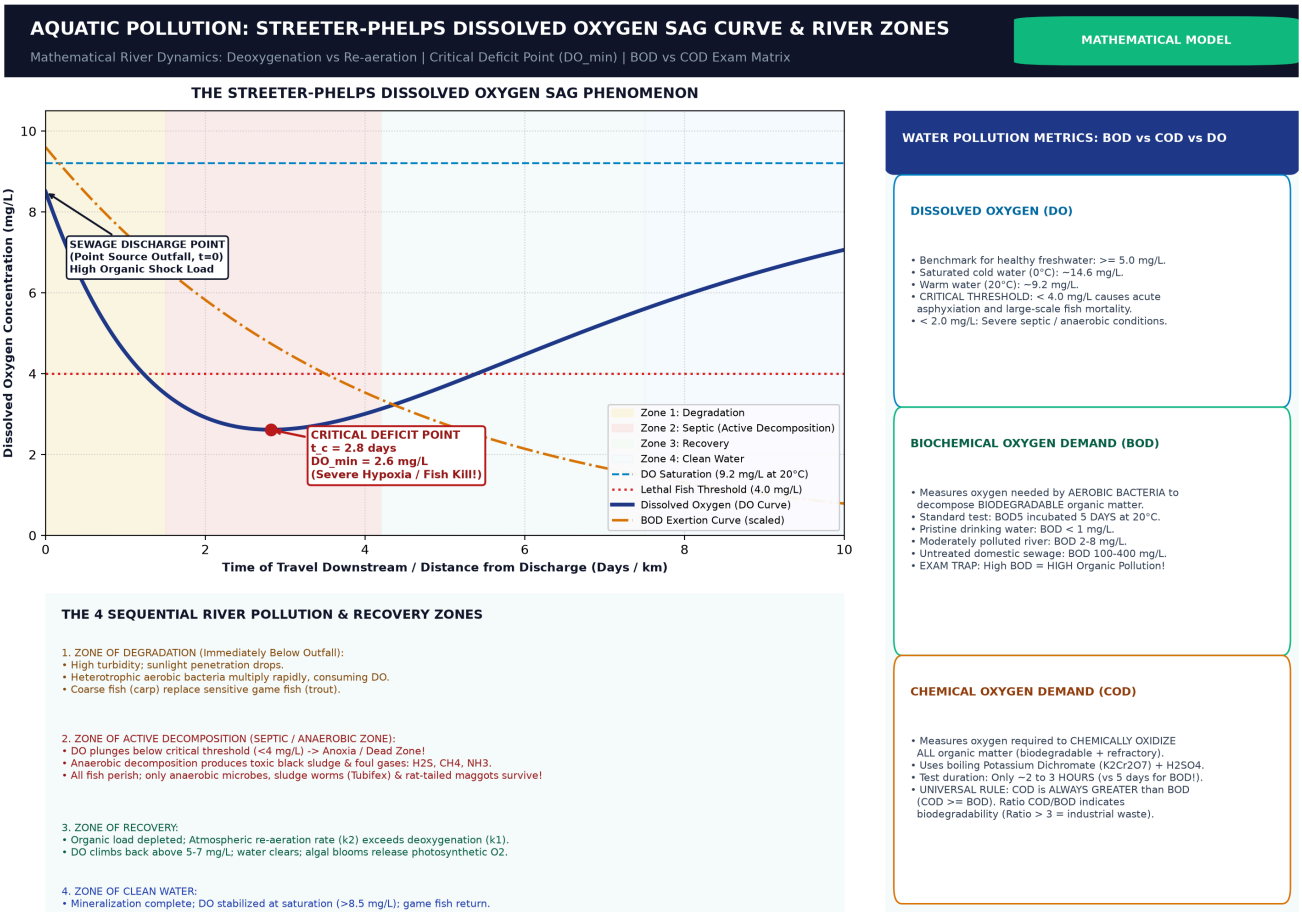
C. Both

D. Only 1

Show answer

Ans: D

Pollutant	CPCB 24-h standard (µg/m³)
PM2.5	60
PM10	100
SO_2	80
NO_2	80



9.4 Water Pollution

Identity: BOD/COD/DO | Eutrophication | Namami Gange (नमामि गंगे) | CGWB trap

- **Water pollution** alters water quality so it becomes unfit for drinking, aquatic life, or irrigation.
- **BOD (Biochemical Oxygen Demand)** measures oxygen consumed by microbes decomposing **organic** matter. **Higher BOD = more organic pollution** (and usually lower DO).
- **COD (Chemical Oxygen Demand)** measures total oxidizable matter. **COD is always ≥ BOD**.
- **DO (Dissolved Oxygen)** above **6 mg/L** suits healthy rivers. Below about **5 mg/L** stresses fish; teaching banks often call **<4 mg/L** highly polluted.
- **Eutrophication** begins when excess **nitrogen and phosphorus** cause algal blooms, bacterial decomposition, and **DO collapse** — a **water** phenomenon, not air pollution.
- **Faecal coliform** bacteria indicate sewage contamination. Safe drinking water should have **zero faecal coliform**.
- Highest **BOD** on the Ganga is classically taught between **Kanpur and Allahabad (Prayagraj)** from heavy pollutant load.
- **Damodar** is called a **biological desert** along the Giridih–Durgapur industrial/mining stretch.
- **Leather industry** effluents rank among the worst chemical polluters of water and soil.
- Disease locks: **arsenic** → Black Foot / cancers (WHO limit **0.01 mg/L**); **fluoride** → fluorosis / **knock-knee**; **mercury** → Minamata; **cadmium** → Itai-itai.
- **Occupational Lung Diseases / Pneumoconiosis (High-Yield UPPCS Match):**

Disease	Causative Pollutant / Dust	Industry / Occupational Source	Key feature / Nickname
Silicosis	Free silica (SiO_2) dust	Stone crushing, quarrying, mining, glass factories	Progressive lung fibrosis; Rajasthan first state to frame Silicosis Policy (2019)
Asbestosis	Asbestos fibers	Roofing sheets, pipe insulation, shipbuilding	Leads to pulmonary fibrosis and **Mesothelioma** (cancer of pleural lining)
Anthracosis	Coal dust	Coal mining (Jharia, Raniganj, Singrauli)	**Black Lung Disease** / Coal Workers' Pneumoconiosis
Byssinosis	Cotton fiber dust	Textile mills, ginning factories	**Brown Lung Disease**; called **"Monday Fever"** (symptoms peak after weekend return)
Bagassosis	Sugarcane bagasse dust	Sugar mills, paper & cardboard manufacturing	Fungal spores (*Thermoactinomyces*) on decomposing crushed sugarcane stalk
Siderosis	Iron / iron oxide dust	Iron foundries, arc welding, iron ore mining	Iron dust accumulation in lung parenchyma
Farmer's Lung	Moldy hay / straw dust	Agriculture, silos, threshing	Hypersensitivity reaction to thermophilic actinomycetes
- Arsenic groundwater hotspots: Ganga–Brahmaputra plains and Bangladesh's Padma–Meghna; Indian states include **West Bengal** (first reported **1978**), Bihar, UP, Jharkhand, Assam, Manipur, and Chhattisgarh.			
- **Kanpur's tanneries** discharge **chromium** into the Ganga as a major UP point source.			
- The **Gomti** through **Lucknow** (लखनऊ) has been flagged among India's most polluted stretches. The **Yamuna** (यमुना) is heavily polluted in the Mathura (मथुरा)–Agra–Delhi stretch.			
- **Namami Gange (2014)** is India's flagship Ganga cleaning mission. **NMCG** is the nodal agency.			
- The **Annual Groundwater Quality Report** is prepared by **CGWB**, not CPCB.			

Disease	Causative Pollutant / Dust	Industry / Occupational Source	Key feature / Nickname
- Rail **bio-toilets** use **anaerobic bacterial** inoculum (DRDO–Railways), producing water and gases after chlorination — not fungal-only, and not ammonia-only end products.			

> **Logic:** — groundwater report = **CGWB only**. Eutrophication ≠ air pollution.

PYQ — UPPCS Prelims 2025, Q70

Annual Groundwater Quality Report 2024 prepared by: 1. CPCB 2. CWC 3. CGWB

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

Show answer

Ans: B

Parameter	Clean river	Polluted river
BOD (mg/L)	<3	>10
DO (mg/L)	>6	<4
pH	6.5–8.5	<6 or >9

9.5 Soil Pollution

Identity: Pesticides | Heavy metals | Acid rain | Kanpur chromium

- **Soil pollution** is the buildup of toxic chemicals, salts, or waste that reduces fertility and enters the food chain (खाद्य श्रृंखला).
- **Organochlorine pesticides** such as **DDT** persist for years and **biomagnify** in food chains.
- **Heavy metals** including **chromium, cadmium, lead, and arsenic** accumulate in topsoil from industry and sludge.
- **Kanpur tanneries** release **hexavalent chromium**, contaminating soil and river sediments in UP.
- **Acid rain** from **SO_2 and NOx** acidifies soil and releases toxic aluminium.
- **Plastic mulch and microplastics** alter soil structure and microbial activity.

- **Phytoremediation** uses plants to absorb contaminants. **Bioremediation** uses microbes to detoxify soil/water by enhancing natural biodegradation.
- Bioremediation **cannot** readily and completely treat many **heavy metals** such as **cadmium and lead**. Genetic engineering can create specialised strains (e.g. *Pseudomonas putida* for oil spills).
- **Oilzapper** (TERI) is a bacterial consortium for oily sludge and oil-spill remediation.
- Indiscriminate fertilisers drive soil, water, and air pollution; residues can enter the food chain. Soil pollution often **leaches into groundwater** over time.

> **Logic:** Soil **erosion** is physical degradation. Chemical contamination = soil **pollution**. Heavy metals ≠ easy bioremediation.

9.6 Noise Pollution

Identity: dB scale | CPCB zone limits | Noise Rules 2000 | Health effects

- **Noise pollution** is unwanted sound that harms hearing, sleep, and health.
- **Decibel (dB)** (डेसिबल) uses a **logarithmic scale**. A **10 dB** rise means **10 times** the sound intensity.
- **Day** is **6 AM–10 PM**. **Night** is **10 PM–6 AM**. Night limits are stricter.
- **CPCB residential limits** are **55 dB (day)** and **45 dB (night)**.
- **Silence zones** near hospitals and schools allow **50 dB (day)** and **40 dB (night)**.
- **Industrial zones** allow **75 dB (day)** and **70 dB (night)**.
- Under **Noise Pollution (Regulation and Control) Rules, 2000**, a complaint may be made when noise exceeds the prescribed ambient standard by **10 dB(A) or more** (Rule 7). Do not key 5 / 8 / 9 dB.
- Prolonged exposure above **85 dB** can cause permanent **Noise-Induced Hearing Loss (NIHL)**.
- Rough dB anchors: rustling leaves **~20 dB**, normal talk **~60 dB**, truck **~80–85 dB**, jet engine **~120 dB**, jet landing **~150 dB**, rocket **~180 dB**.
- Noise from aircraft faster than sound is a **sonic boom** (Mach scale: Mach-1 = speed of sound).
- **Green muffler** means rows of trees/shrubs planted to absorb noise (and dust) near roads and industry.
- **Moradabad (UP)** ranked among the world's most noise-polluted cities in UNEP's Frontier Report 2022.
- **Noise Pollution (Regulation and Control) Rules, 2000** operate under **EPA 1986**. Jet flights are among the most intense common noise sources.

> **Logic:** Complaint threshold = **+10 dB(A)** above ambient, not 5/8/9. **45 dB** is residential **night**, not silence zone night (**40 dB**). Green muffler = noise control, not soil.

9.7 Radioactive Pollution

Identity: Ionizing radiation | Bq/Sv | Half-life | AERB

- **Radioactive pollution** comes from ionizing radiation exceeding safe dose limits.
- **Becquerel (Bq)** measures radioactivity as disintegrations per second.
- **Sievert (Sv)** measures biological radiation dose.
- **Half-life** is the time for **50%** of a radioactive isotope to decay.
- **Natural sources** include cosmic rays and **radon gas** in buildings.
- **Anthropogenic sources** include nuclear power plants, medical isotopes, and uranium mining.
- **Chernobyl (26 April 1986, Ukraine / then USSR)** is the classic nuclear accident case for radioactive pollution teaching.
- Radioactive pollution can cause **hereditary damage**, blood-circulation harm, and **carcinogenesis** — it is not mainly a soil-mineral imbalance story.
- **Bhopal Gas Tragedy (2–3 December 1984)** at Union Carbide released **methyl isocyanate (MIC, CH_3NCO)** — India's worst industrial gas disaster.
- **Alpha particles** are dangerous if ingested. **Gamma rays** penetrate the body from outside.
- **AERB (Atomic Energy Regulatory Board)** regulates nuclear safety in India.
- Radioactive biomedical waste needs **shielded containers** under **BMW Rules 2016**.

> **Logic:** Half-life means **50% decay**, not complete safety. Bhopal = MIC 1984, not CO/SO_2.

9.8 Thermal Pollution

Identity: Heated effluent | ↓DO | Power plants | Cooling towers

- **Thermal pollution** raises water temperature above the natural range, mainly from industrial cooling water.
- **Warm water holds less dissolved oxygen**. Fish need more oxygen as metabolism rises with heat.
- Even a **2–3°C** rise can stress aquatic ecosystems.
- **Coal and nuclear power plants** are major sources of thermal discharge.
- **Cooling towers** and **cooling ponds** reduce thermal pollution before release.
- Thermal pollution is regulated through **consent conditions under the Water Act 1974**.

> **Logic:** Fish kills mainly follow **DO reduction**, not direct heat toxicity alone.

9.9 Marine Pollution

Identity: 80% land-based | MARPOL | Oil/plastic/nutrients | Coral bleaching (प्रवाल विरंजन)

- **Marine pollution** contaminates oceans through sewage, oil, plastics, and nutrients.

- About **80% of marine pollution originates on land** via rivers and coastal discharge.
- **Oil spills** form slicks that block oxygen exchange and coat wildlife.
- **Plastic waste** and **microplastics** enter marine food chains globally.
- **Nutrient runoff**, estuary (ज्वारनदमुख) discharges, monsoon run-off, and coastal **upwelling** can all drive harmful **algal blooms**.
- **Ocean acidification** is falling ocean pH from atmospheric **CO_2** uptake; it harms **calcareous phytoplankton**, **coral reefs**, and some animals with phytoplankton larvae (cloud seeding is **not** the usual acidification concern).
- Oceans absorb about **one-third** of anthropogenic CO_2.
- **MARPOL 73/78** is the international convention to prevent pollution from ships (नावाध्यक्ष). The **International Maritime Organization (IMO)** HQ is in **London** (IMCO renamed IMO in **1982**).
- **SACEP** = **South Asia Cooperative Environment Programme** (oil/chemical pollution MoU framing in papers).
- **MARPOL Annex V** prohibits plastic discharge from vessels.
- **Coral bleaching** in the Andaman (अंडमान) and Nicobar (निकोबार) Islands was linked to **Indian Ocean warming of about 2°C** (2024 PYQ pattern).
- **Oilzapper**-type bioremediation has been used for oily sludge/spills (e.g. Mumbai port (बंदरगाह) spill teaching note).

> **Logic:** — bleaching from **temperature rise**, not temperature decrease. Acidification concern ≠ cloud seeding.

PYQ — UPPCS Prelims 2024, Q19

Coral bleaching in Andaman & Nicobar caused by:

A. Increased coastal industry

B. **+2°C Indian Ocean warming**

C. Heavy marine pollution

D. −4°C temperature drop

Show answer

Ans: B — Warming is the primary trigger; pollution is a secondary stressor.

9.10 Plastic Pollution

Identity: PWM Rules 2016 | SUP ban 2022 | Microplastics | EPR

- **Plastic pollution** persists because polymers take **100–1000 years** to break down.
- **Polythene** is a polymer of **ethylene (C_2H_4)** — carbon–hydrogen chains that do not biodegrade readily.

- **Microplastics** are fragments **smaller than 5 mm**. **Microbeads** (often polyethylene in cosmetics) harm **marine ecosystems**.
- **Plastic Waste Management Rules, 2016** regulate manufacture, recycling, and **EPR**.
- The **SUP (single-use plastic) ban notification of July 2022** banned identified items such as plastic cutlery, straws, and certain films.
- **EPR** makes producers and brand owners collect equivalent plastic waste they place in the market; EPR featured early in **Plastic/E-waste Rules 2011**.
- **India is the 3rd largest plastic waste generator** globally after the USA and China.
- World Environment Day (विश्व पर्यावरण दिवस) **2023** theme focused on solutions to plastic pollution (**Beat Plastic Pollution**).
- The **Ganga** (गंगा) carries large amounts of plastic waste from ghats and urban dumping.
- **MARPOL Annex V** bans plastic discharge from ships into the sea.
- **Incinerators** burn waste. **Pyrolysis / plasma gasification** are waste-to-energy thermal routes.
- **R2 Code of Practices** promotes environmentally responsible **electronics recycling**.

> **Logic:** PWM Rules = **2016**. **SUP ban = 2022**. Microbeads = marine harm, not mainly skin cancer.

9.11 Solid Waste Management

Identity: SWM Rules 2016 | 3R hierarchy | Segregation | Swachh Bharat

- **Solid waste management** covers collection, segregation, treatment, and disposal of municipal and commercial waste.
- The **3R hierarchy** is **Reduce > Reuse > Recycle > Recovery > Disposal**.
- **Solid Waste Management Rules, 2016** make **segregation at source** mandatory into wet, dry, and domestic hazardous fractions.
- The Rules provide **exact and elaborate criteria** for identifying landfill and waste-processing sites.
- **Wet waste** is treated by **composting** or **biomethanation**.
- **Dry waste** goes to **Material Recovery Facilities (MRF)** for recycling.
- **Landfill** is the **last resort** for inert rejects only, in lined sites.
- **Swachh Bharat Mission (2014)** targeted open defecation free status and waste management. **SBM 2.0 (2021–2026)** targets garbage-free cities.
- Biomedical, hazardous, and e-waste follow **separate rules**, not SWM Rules alone.

> **Logic:** SWM Rules = **2016**. Segregation is **mandatory**, not optional.

9.12 Biomedical Waste

Identity: BMW Rules 2016 | Colour coding | CBWTF | 48-hour limit

- **Biomedical waste (BMW)** comes from hospitals, clinics, labs, and veterinary institutions.
- **BMW Rules 2016** replaced the 1998 rules and require separate collection from municipal waste.
- **Yellow bags** hold anatomical waste, soiled dressings, and expired medicines for **incineration or autoclaving**.
- **Red bags** hold contaminated recyclable disposables for **disinfection then recycling**. Classic Red list items include **IV tubes and sets**, **catheters**, **urine bags**, and **syringes without needles**.
- **White translucent containers** hold **sharps** such as **needles** from used syringes and scalpels.
- **Blue bags** hold **glassware** (including contaminated glass vials) and metallic implants after disinfection.
- **CBWTF (Common Biomedical Waste Treatment Facility)** serves clusters of healthcare units.
- Untreated BMW must not be stored beyond **48 hours**.

> **Logic:** Sharps = **white**, not yellow. Needles are **not** Red. Contaminated glass vials are **Blue**, not Red. Yellow = anatomical/infectious soiled waste.

Colour	Waste type	Treatment
Yellow	Anatomical, soiled, medicines	Incineration/autoclave
Red	Contaminated recyclables (IV tubes, catheters, urine bags, syringes without needles)	Disinfection → recycle
White	Sharps (needles)	Autoclave → disposal
Blue	Glass vials, metal	Disinfection → recycle

9.13 Hazardous Waste

Identity: HW Rules 2016 | Basel Convention | 90-day storage | Kanpur chromium sludge

- **Hazardous waste** threatens health or the environment because it is toxic, reactive, corrosive, or flammable.
- **Hazardous and Other Wastes Rules, 2016** list generating processes in **Schedule I** and hazardous constituents in **Schedule II**.
- Hazard characteristics include **ignitability, corrosivity, reactivity, and toxicity**.
- Storage is allowed up to **90 days** unless an extension is granted.
- The **manifest system** tracks waste from generator to transporter to treatment facility.
- The **Basel Convention (1989)** (बासेल कन्वेंशन) controls **transboundary movement** of hazardous waste. India ratified it in **1992**.
- **Kanpur tannery chromium sludge** is a classic UP hazardous waste example.

> **Logic:** Basel = **hazardous waste movement**, not climate or biodiversity.

9.14 E-waste

Identity: E-waste Rules 2022 | 3rd largest generator | EPR targets | Lead/mercury/cadmium

- **E-waste** is discarded electrical and electronic equipment and its components.
- India is the **3rd largest e-waste generator** globally after China and the USA.
- **E-waste (Management) Rules, 2022** replaced the 2016 rules and strengthened **EPR**.
- **Schedule I** covers **67 categories** of EEE from phones and computers to refrigerators and lamps.
- Hazardous components include **lead** (CRT screens), **mercury** (CFLs), and **cadmium** (batteries).
- Valuable metals such as **gold, copper, and palladium** can be recovered from circuit boards.
- **EPR collection targets** for producers rise from about **60% toward 80%** of market placement.
- Consumers must channel e-waste only to **authorized collectors and recyclers**.
- **Moradabad and Noida** region has significant e-waste generation and informal processing in UP.

> **Logic:** Current rules = **2022**. India rank = **3rd**, not 1st.